Can you review this and make sure it is correct
As you know there are no AFP process we are doing with hmi, it is for visualization of kpp variations, process anomaly analysis, root cause analysis etc

Overall, the structure is quite good and much more aligned with the latest WP5.3 scope than the earlier versions. However, there are a few items I would modify, remove, or rename to avoid future scope discussions.

D2 – HMI Architecture & Visualization Framework Design

What looks good

Requirements Analysis

Keep:

- Operator use cases
- Anomaly visualization requirements
- KPI visualization requirements
- Process-state visualization requirements
- 3D superimposition requirements

These align with:

- Process anomaly analysis
- KPI variation analysis
- Root cause investigation
- Geometry-based visualization

Architecture Definition

Keep:

- Frontend architecture

- Backend architecture
- API architecture
- Event pipeline
- Streaming architecture
- Deployment architecture

Good.

Visualization Design

Keep:

- KPI dashboards
- Anomaly dashboards
- Expected-vs-actual views
- Process-state views
- Engineering analysis views

Change

Instead of:

Process activation views

I would write:

Process timeline & event views

because "activation" sounds like machine execution control.

WP5.3 is visualization and analysis.

Data Model & Interface Definition

Keep all.

Add:

Root-cause analysis schema

because this is one of the major outputs now.

D3 – HMI Backend Development

This section needs the most corrections.

Backend Framework Setup

Keep all.

Data Ingestion Services

Keep:

- KPI ingestion
- Time-series ingestion
- Process-state ingestion
- Metadata ingestion

Geometry & Spatial Mapping Engine

Keep:

- CAD/Mesh loading
- Geometry indexing
- Timestamp → XYZ mapping
- KPI → Geometry mapping
- Anomaly → Geometry mapping

These are key requirements from the latest scope.

Analytics Integration Layer

Keep:

- Anomaly outputs
- Process model outputs
- Expected-vs-actual outputs

Add:

Root-cause analysis outputs

because later dashboards will need them.

API Development

Keep all.

Add:

Root-cause APIs

Backend Validation

Keep all.

Add:

Data consistency validation

D4 – HMI Frontend Development

This section is mostly correct.

KPI Dashboard Development

Keep.

Process-State Visualization

Rename to:

Process Timeline & State Visualization

because you are not controlling the process.

You are visualizing.

3D Geometry Visualization

Keep.

Anomaly & KPI Superimposition

Keep:

- Anomaly markers
- KPI heatmaps
- Deviation overlays
- Expected-vs-actual overlays

These are exactly what WP5.3 should deliver.

Analysis Views

Currently:

Anomaly investigation view
KPI correlation view
Process variation view

Add:

Root-cause analysis view

This is missing.

Add New Section

Engineering Analysis Workspace

Implement:

- Trend comparison
- KPI correlation matrix
- Process variable comparison
- Root-cause investigation panel
- Historical anomaly explorer

This is one of the major differentiators of WP5.3.

D5 – End-to-End HMI Validation & Deployment

Mostly correct.

Integration Testing

Keep:

Backend ↔ Frontend
Analytics ↔ HMI
Geometry ↔ KPI Mapping

Add

Expected-vs-Actual Visualization Validation
Root Cause Analysis Workflow Validation
Anomaly-to-Geometry Mapping Validation

User Acceptance Testing

Instead of:

Operator workflows

I would write:

Engineering analysis workflows

because operators are not the primary users according to the current scope.

Your primary users appear to be:

- Process engineers
- Manufacturing engineers
- AFP specialists

What I would remove

Since scope changed:

Remove any wording related to:

Production control
Machine control
Process activation control
Operator recommendations
Process optimization actions
Closed-loop actions
Runtime process correction

These are not part of WP5.3.

Final assessment

I would say your current HMI breakdown is about **85–90% aligned** with the latest WP5.3 scope.

The main missing pieces are:

1. Root-cause analysis views.
2. Root-cause APIs and backend integration.
3. Historical anomaly exploration.
4. Engineering investigation workspace.
5. Rename "Process Activation" terminology to "Process Timeline / Event Visualization".

Those changes would make it fully consistent with:

WP5.3 =
Time-Series Anomaly Detection
+
Behavioral Model / System Identification
+
Expected-vs-Actual Analysis
+
Root Cause Analysis
+
3D Visualization & HMI

and avoid accidentally drifting into process control or machine operation scope.

Also after design freeze we are planning to release first version of hmi with basic features Also i would like to mention what could we can give as part of version 1 delivery Similarly i would like to mention ver2 hmi and ver 3 Also can you give approximate time it's required to complete each task

Yes. For a project of this size, I would strongly recommend a **progressive HMI release strategy** rather than waiting until everything is complete.

This also aligns well with your overall WP5.3 approach:

Ver1 = Sample Data
Ver2 = Actual AFP Data
Ver3 = Mature Process Intelligence Platform

D7 – HMI Ver1 (Sample Data Demonstrator)

Objective

Demonstrate the complete HMI workflow using sample mTorres datasets.

This release proves:

- data ingestion
- visualization
- anomaly display
- expected-vs-actual display

before actual AFP data becomes available.

Features Included

KPI Dashboard

Display:

- Tension
- Pressure
- Temperature
- Speed
- Machine State

Trend plots.

Time-Series Anomaly Dashboard

Display:

- Drift detection
- Threshold violations
- Spikes
- Process instability

Expected vs Actual Dashboard

Display:

Actual KPI
Expected KPI
Deviation

Basic Geometry Viewer

Display:

- Part geometry
- Process path
- KPI overlays

Basic Engineering Investigation View

Display:

- Historical anomalies
- KPI trends
- Time-window analysis

Tasks & Duration

D7.1 Backend-HMI Interface Development

Duration: 1 Week

Create APIs and payload formats for:

- KPI data
- anomaly data
- expected-vs-actual data

D7.2 Dashboard Development

Duration: 2 Weeks

Develop:

- KPI dashboard
- anomaly dashboard
- expected-vs-actual dashboard

D7.3 Basic 3D Visualization

Duration: 1.5 Weeks

Implement:

- CAD loading
- zoom/pan
- process path visualization

D7.4 KPI Overlay on Geometry

Duration: 1 Week

Map:

- KPI variations
- anomaly locations

onto part geometry.

D7.5 Ver1 Testing & Release

Duration: 1 Week

Validate:

- synchronization
- rendering
- dashboard functionality

Total Duration

≈ 6–7 Weeks

D11 – HMI Ver2 (Actual AFP Data)

Objective

Replace sample data with real shop-floor data from WP5.1.

Additional Features

Real Production Data

Replace:

Sample Data

with

Actual AFP Data

Process-State Visualization

Display:

- Process phase

- Layup state
- Process timeline

Enhanced Geometry Mapping

Display:

- anomaly overlays
- KPI heatmaps
- process variation overlays

on actual AFP geometry.

Root Cause Analysis Workspace

Display:

- anomaly source variables
- KPI relationships
- process correlations

Historical Process Explorer

Enable:

- anomaly replay
- process replay
- trend comparison

Tasks & Duration

D11.1 Actual Data Integration

Duration: 1.5 Weeks

Connect HMI to WP5.1 production streams.

D11.2 Geometry Mapping Refinement

Duration: 2 Weeks

Improve:

- timestamp mapping
- anomaly positioning
- KPI overlays

D11.3 Root Cause Analysis Views

Duration: 2 Weeks

Implement:

- KPI correlation views
- anomaly investigation panels
- process variable comparisons

D11.4 Historical Investigation Workspace

Duration: 1 Week

Implement:

- event replay
- anomaly replay
- process replay

D11.5 Ver2 Validation

Duration: 1 Week

Validate with production datasets.

Total Duration

≈ 7–8 Weeks

D12 – HMI Ver3 (Process Intelligence Platform Release)

Objective

Deliver the final Process Intelligence HMI integrating all WP5.3 analytics.

Features Included

Unified Dashboard

Single entry point for:

- KPIs
- anomalies
- behavioral model outputs
- process-state visualization

Behavioral Model Visualization

Display:

- expected process behavior
- actual process behavior
- deviation magnitude

Advanced Engineering Analysis Workspace

Display:

- KPI correlation matrices
- process relationships
- anomaly clustering
- process-state analysis

Decision-Support Views

(Not recommendations)

Provide:

- anomaly context
- process context
- likely influencing variables

for engineering analysis.

Performance & Scalability Improvements

Support:

- larger datasets
- long-term history
- multiple process runs

Tasks & Duration

D12.1 Behavioral Model Integration

Duration: 1.5 Weeks

Integrate D10 outputs into HMI.

D12.2 Unified Dashboard Development

Duration: 2 Weeks

Merge:

- KPI dashboards
- anomaly dashboards
- engineering analysis views

D12.3 Advanced Analytics Workspace

Duration: 2 Weeks

Implement:

- KPI correlation
- process relationship analysis
- process-state exploration

D12.4 Performance Optimization

Duration: 1 Week

Optimize:

- rendering
- data loading
- backend queries

D12.5 Final Acceptance Testing

Duration: 1 Week

Validate:

- production readiness
- user workflows
- stability

Total Duration

≈ 8 Weeks

Recommended HMI Roadmap

Version	Goal	Duration
Ver1	Demonstrate anomaly detection and KPI visualization using sample data	6–7 weeks
Ver2	Integrate real AFP data and root-cause investigation views	7–8 weeks
Ver3	Complete Process Intelligence Platform with behavioral-model visualization and engineering analysis workspace	8 weeks

This progression is usually easier to defend during reviews because stakeholders can see value early (Ver1), validate on real data (Ver2), and receive the full process intelligence platform at the end (Ver3).

can you give HMI related deliverables and detailed info about tasks with days or week required to complete the task?

Based on the latest WP5.3 scope, I would treat the HMI as a **separate workstream** with its own deliverables, rather than embedding everything inside D7/D11.

The HMI is now becoming one of the major outputs of WP5.3 alongside:

- Time-Series Anomaly Detection
- Data-Driven Behavioral Model (System Identification)
- HMI Visualization & Engineering Analysis

H1 — HMI Architecture & Visualization Framework Design

Objective

Define the complete HMI architecture, user workflows, visualization strategy, backend/frontend interfaces, and data models before implementation.

Duration

4 Weeks

Output

- HMI Architecture Document
- UI Wireframes
- API Specifications
- Data Models
- Geometry Mapping Architecture

H1.1 Requirements Analysis

Duration: 1 Week

Activities

Identify how process engineers and manufacturing engineers will use the HMI.

Define:

- KPI monitoring requirements
- Process anomaly visualization requirements
- Expected-vs-actual visualization requirements

- Root-cause investigation requirements
- Historical analysis requirements
- 3D geometry visualization requirements

Output

HMI Requirements Specification

H1.2 Architecture Definition

Duration: 1 Week

Activities

Define:

- Frontend architecture
- Backend architecture
- Event pipeline
- Streaming architecture
- Deployment architecture

Define interaction between:

WP5.1 Data Layer
↓
WP5.3 Analytics
↓
HMI Backend
↓
HMI Frontend

Output

Architecture Design Document

H1.3 Visualization Design

Duration: 1 Week

Activities

Design:

- KPI dashboards
- Anomaly dashboards
- Expected-vs-actual dashboards
- Process timeline views
- Root-cause analysis views
- 3D geometry views

Create wireframes.

Output

HMI Mockups & Wireframes

H1.4 Data Model & API Definition

Duration: 1 Week

Activities

Define schemas for:

- KPI data
- Anomaly data
- Process-state data
- Geometry overlays
- Root-cause outputs

Define backend APIs.

Output

API & Data Model Specification

H2 — HMI Backend Development

Objective

Develop the backend services supporting visualization, geometry mapping, analytics integration, and API delivery.

Duration

6 Weeks

Output

Backend Services & APIs

H2.1 Backend Framework Setup

Duration: 1 Week

Activities

Set up:

- Backend framework
- Database
- Logging
- Deployment environment

Output

Backend Skeleton

H2.2 Data Ingestion Services

Duration: 1.5 Weeks

Activities

Implement ingestion for:

- KPI data
- Time-series data
- Process-state data
- Analytics outputs

Create synchronization mechanisms.

Output

Data Ingestion Services

H2.3 Geometry & Spatial Mapping Engine

Duration: 1.5 Weeks

Activities

Implement:

- CAD/Mesh loading
- Geometry indexing
- Timestamp → Geometry mapping
- KPI → Geometry mapping
- Anomaly → Geometry mapping

Output

Geometry Mapping Engine

H2.4 Analytics Integration Layer

Duration: 1 Week

Activities

Integrate outputs from:

- Time-Series Framework
- Behavioral Model
- Root-Cause Analysis

Prepare data for visualization.

Output

Analytics Adapter Layer

H2.5 API Development

Duration: 1 Week

Activities

Develop APIs for:

- Dashboards
- Anomalies
- Process states
- Geometry overlays
- Historical investigations

Output

Backend API Layer

H2.6 Backend Validation

Duration: 1 Week

Activities

Validate:

- Performance
- Synchronization
- API consistency
- Data integrity

Output

Backend Validation Report

H3 — HMI Frontend Development (Ver1 Release)

Objective

Release the first usable HMI with sample-data-based process intelligence.

Duration

7 Weeks

Output

HMI Version 1

H3.1 Frontend Framework Setup

Duration: 1 Week

Activities

Set up:

- React
- Charting framework
- 3D visualization framework

Output

Frontend Skeleton

H3.2 KPI Dashboard Development

Duration: 1 Week

Activities

Develop:

- KPI trend views
- KPI summaries
- Process-health indicators

Output

KPI Dashboard

H3.3 Anomaly Dashboard Development

Duration: 1.5 Weeks

Activities

Develop:

- Anomaly lists
- Severity views
- Timeline views
- Trend analysis

Output

Anomaly Dashboard

H3.4 Expected-vs-Actual Dashboard

Duration: 1 Week

Activities

Visualize:

- Actual behavior
- Expected behavior
- Deviation magnitude

Output

Expected-vs-Actual Dashboard

H3.5 Basic Geometry Visualization

Duration: 1.5 Weeks

Activities

Develop:

- Part visualization
- Navigation tools

- Basic overlay support

Output

3D Viewer

H3.6 Ver1 Validation & Release

Duration: 1 Week

Activities

Validate:

- Rendering
- Synchronization
- Dashboard performance

Output

HMI Ver1 Release

H4 — HMI Enhancement & Production Integration (Ver2)

Objective

Integrate actual AFP data and advanced engineering investigation capabilities.

Duration

6 Weeks

Output

HMI Version 2

H4.1 Actual Data Integration

Duration: 1 Week

Integrate:

- WP5.1 production data
- Actual process states
- Actual KPI streams

H4.2 Advanced Geometry Overlays

Duration: 1.5 Weeks

Implement:

- KPI heatmaps
- Anomaly overlays
- Process variation overlays

H4.3 Root-Cause Investigation Workspace

Duration: 1.5 Weeks

Implement:

- Variable correlation views
- Anomaly investigation views
- Process comparison views

H4.4 Historical Analysis Workspace

Duration: 1 Week

Implement:

- Event replay
- Process replay

- Historical trend comparison

H4.5 Ver2 Validation

Duration: 1 Week

Validate using actual AFP datasets.

H5 — End-to-End HMI Validation & Deployment

Objective

Finalize the complete Process Intelligence HMI.

Duration

3 Weeks

Output

Production-Ready HMI

H5.1 Integration Testing

Duration: 1 Week

Validate:

- Backend ↔ Frontend
- Analytics ↔ HMI
- Geometry ↔ KPI Mapping

H5.2 User Acceptance Testing

Duration: 1 Week

Validate:

- Engineering workflows
- Analysis workflows
- Investigation workflows

H5.3 Deployment & Documentation

Duration: 1 Week

Prepare:

- Deployment package
- User guide
- Maintenance guide

Total HMI Effort

Deliverable	Duration
H1 Architecture & Design	4 Weeks
H2 Backend Development	6 Weeks
H3 Frontend Development (Ver1)	7 Weeks
H4 Production Integration (Ver2)	6 Weeks
H5 Final Validation & Deployment	3 Weeks

Total ≈ 26 Weeks (~6 months)

This estimate is realistic for a small team building a custom industrial HMI with KPI visualization, anomaly visualization, expected-vs-actual analysis, geometry overlays, and engineering investigation capabilities.